

Latency-Budget Reference Card

Day 4 · Activity 3 handout. Take your latency SLO and walk it across the Container diagram. Each hop gets a budget; the sum must fit the SLO. If it doesn't, something has to give — surface that math *before* the feature ships and falls short.

Per-hop estimates (use these to annotate each arrow)

Hop type	Typical range
Network hop (LAN → internet RTT)	20–80 ms
Database read (indexed)	5–30 ms
Database write (single-row)	10–50 ms
External API call (synchronous)	50–500 ms — wildly varies; look up real numbers
Async publish (Event Hubs / Service Bus, local)	1–10 ms

Async publishes are cheap on the hot path — but they don't count against user-visible latency the way a synchronous external call does. The **synchronous external call** is the hop teams most often under-estimate.

The per-hop walk

If a feature has a top-level SLO of **p95 ≤ 400ms**, that 400ms must be **divided across the hops**:

```
[Mobile] →(80ms net)→ [API gateway] →(20ms)→
[Reconcile module]

                                ↓
                                (60ms)
[Postgres read]

                                ↓
                                (20ms)
[Tickets module call]

                                ↓
                                (40ms)
[Postgres write]

                                ↓
                                (40ms)
[Audit publish]

                                ↓
                                Server-side:

20+60+20+40+40 = 180ms                                Network: 80ms

each way x2 = 160ms                                Total ≈ 340ms

(under 400ms p95)
```

If the math doesn't fit — what gives

- **Tighter SLO target** (less ambitious).
- **Reduce hops** (cache, async, denormalize).
- **Faster individual hops** (engineering work).
- **Wider SLO percentile** (e.g., relax p99 → p95).

The **wider-percentile** option is the most overlooked. Don't reach for p99 if you have no instrumentation — p95 is usually fine for B2B SaaS.

Worked example — FieldPulse reconcile (SLO: p95 ≤ 1000ms)

Path: Mobile → API GW → Reconcile module → Postgres read
→ Tickets
module
→ Postgres
write
→ Audit
publish
→ response back to Mobile

Mobile→API GW round trip:	80ms
API GW → Reconcile module:	20ms
Postgres read (current state):	30ms
Tickets module call (sync):	100ms (their median)
Postgres write:	40ms
Audit publish (async):	10ms
Reconcile → API GW response:	20ms
API GW → Mobile response:	80ms
	~380ms median
	p95 likely ~700–900ms
	✓ Fits the 1000ms SLO

Budget remaining for variance: ~100–300ms
Risk: Tickets module call dominates; if their latency degrades, we breach.

Your annotation table

Arrow / hop	Type	Estimate (ms)
Total (server + net round trip)		

- SLO target: _____ ms · Sum vs SLO: fits / doesn't fit
- If it doesn't fit — what gives:

- **Question for the architect:**
